

Exact Certificate Complexity versus Intrinsic Support in Quantum Compilation

Paper B — hardened manuscript V2 Scientific cut: Paper-B/B1 parents plus R2 alphabet-Davenport theorem Workflow cut: academic-paper-skills@188e83e639571c435344630ae68fdc66072650d2

Abstract

A finite-support certificate can be perfectly sharp for its proof language and arbitrarily loose as a description of the compiler it certifies. We formalize this separation using an alphabet-restricted zero-sum invariant. Let H be a finite abelian signature group and $A \subseteq H$ an allowed alphabet. Define $\mathbf{zsf}(H; A)$ as the maximum length of a zero-sum-free word over A . In the abstract deletion proof system whose legal step removes a nonempty proper zero-sum subword from a nonzero-total word, the exact uniform terminal complexity is $\mathbf{zsf}(H; A)$: every longer word is reducible, while a maximum zero-sum-free word is a matching terminal witness. For a production compiler, this remains an exact certificate lower bound only when that witness is realizable.

Two quantum-compilation families provide opposite controls. In one-Tag R6M, the production signature alphabet realizes F_2^2 , the rank-only certificate complexity is two, and independent compiler upper/lower results give intrinsic support $\kappa_{R6M}=2$. In R6I, both production alphabets realize basis obstructions in F_2^5 , so the rank-only certificate complexity is exactly five, while an independent whole-system Tag-relocation theorem gives intrinsic support exactly one. Hence

$$\beta_{\text{rank-only}}(R6I)=5 > 1=\kappa_{R6I}.$$

For t registered independent components, certificate and intrinsic budgets are $5t$ and t ; the additive gap is $4t$, and a direct support enumerator incurs an asymptotic search-volume ratio $\Theta(n^{4t})$ for fixed local alphabet and t . The result is not an unrestricted proof lower bound. It establishes a precise reporting principle: a support number belongs to the compiler only after an intrinsic lower witness; otherwise it is a property of a normalization or certificate language.

1. Introduction

Support bounds make otherwise unbounded exact compiler searches finite. Their numerical value, however, can answer three different questions:

1. What support does the compiler intrinsically require?
2. What support can a named normalization reach?
3. What support can a restricted proof system certify?

These questions coincide in some families and diverge sharply in others. Paper A provides the positive normal-form theorem. This paper identifies its exact proof-language complexity and compares it with independently established intrinsic support.

The central refinement is to replace ambient rank by $\mathbf{zsf}(H; A)$, the longest zero-sum-free word over the actual signature alphabet. In elementary binary systems this equals dimension when a basis is present, but the general definition makes the quantifiers and lower-witness obligation explicit.

1.1 Contributions

1. **Exact abstract certificate theorem:** the zero-sum deletion language has terminal complexity $\mathbf{zsf}(H; A)$.

2. **Realization criterion:** a compiler inherits the matching lower bound only if it realizes a maximum zero-sum-free word.
3. **Tight control:** R6M has certificate and intrinsic support both equal to two.
4. **Strict separation:** R6I has exact rank-only certificate five and intrinsic support one.
5. **Product/search amplification:** registered products give $5t$ versus t and direct-enumeration ratio $\Theta(n^{(4t)})$.

2. Three support quantities

Fix a compiler family F and objective C .

2.1 Intrinsic support

$\kappa(F, C)$ is the least k such that every instance has an exact optimum of support at most k , with an independent witness showing $k-1$ cannot suffice.

2.2 Normalization ceiling

A transformation N may prove every configuration has a no-more-expensive representative of support at most B_N . Without a matching compiler lower witness, B_N belongs to (F, C, N) rather than intrinsically to (F, C) .

2.3 Certificate complexity

A proof system P restricts visible information and legal inference. $\beta_P(F, C)$ is the least uniform support budget that P can certify for its production scope. Soundness gives

$$\kappa(F, C) \leq \beta_P(F, C),$$

but equality is additional mathematics.

3. Alphabet-restricted zero-sum deletion

Let H be a finite abelian group and $A \subseteq H$. A certificate word $v_1 \dots v_w$ has nonzero total. The only legal shortening removes a nonempty proper subword summing to zero.

Define

$$\text{zsf}(H; A) = \max\{|W| : W \text{ is a zero-sum-free word over } A\}.$$

Theorem 1 (exact abstract certificate complexity). The maximum terminal length of the abstract deletion language over all nonzero-total words on A is exactly $\text{zsf}(H; A)$.

Proof. A word longer than zsf contains a nonempty zero-sum subword. Its nonzero total prevents that subword from being the whole word, so a legal deletion exists. Conversely, a maximum zero-sum-free word has nonzero total and admits no deletion.

This theorem is generic zero-sum mathematics. The compiler question begins with realization.

Corollary 2 (production realization). If every production word lies over A , the certificate ceiling is at most $\text{zsf}(H; A)$. If a production state realizes a maximum zero-sum-free word and no other certificate rule can reduce it, the ceiling is exact.

For $H = \mathbb{F}_2^d$, $\text{zsf} \leq d$; any alphabet containing a basis has $\text{zsf} = d$.

4. Tight control: R6M

A one-Tag R6M frame coordinate carries partner and Tag bits. The production alphabet realizes a basis of F_2^2 ; hence the rank-only certificate language has

$$\text{beta_rank-only}(\text{R6M})=2.$$

Independent all-size normalization and exact support-one obstruction results establish

$$\text{kappa_R6M}=2.$$

Therefore R6M is a tight control:

$$\text{beta_rank-only}(\text{R6M})=\text{kappa_R6M}=2.$$

The existence of this control matters. It shows that the certificate is not generically loose; tightness depends on whether its obstruction is also an intrinsic compiler obstruction.

5. Strict separation: R6I

R6I has two rank-2 dependent-triple blocks with a shared two-bit Tag. The production block-deletion alphabets span five-dimensional binary quotients and contain realized basis words. The abstract theorem and production binding give

$$\text{beta_rank-only}(\text{R6I})=5.$$

A stronger transformation leaves the rank-only language. It localizes each block to an anticommuting core and then relocates/reconstructs the shared Tag globally. That all-size theorem proves support at most one, while support zero is infeasible:

$$\text{kappa_R6I}=1.$$

Thus the exact single-component separation is

$$\text{beta_rank-only-kappa}=4.$$

There is no contradiction. The basis word blocks zero-XOR deletion while the successful proof changes auxiliary Tag structure that the certificate freezes.

6. Product amplification

Let F_{R6I}^t be the registered product of t independent components on disjoint coordinates with additive support budget and no cross-component move.

Theorem 3. For every $t \geq 1$,

$$\text{beta_rank-only}(F_{\text{R6I}}^t)=5t,$$

$$\text{kappa}(F_{\text{R6I}}^t)=t.$$

Proof. Componentwise upper bounds add. A realized basis obstruction in each component gives the certificate lower bound five per component. Support zero is infeasible in each component and the support-one normalization acts independently, giving intrinsic lower and upper value one per component.

The additive gap is $4t$.

6.1 Direct enumeration consequence

For fixed local alphabet and fixed support budget B , a direct support enumerator over n coordinates has leading search growth $\Theta(n^B)$. Consequently the registered product's rank-only and intrinsic direct enumerators scale as

$\Theta(n^{(5t)})$ and $\Theta(n^t)$,

with ratio

$\Theta(n^{(4t)})$.

This is an exact statement about the declared enumeration model, not an algorithm-independent complexity-class lower bound.

7. Relation to prior work

General sparse-optimum theory already studies support bounds and lower constructions for integer optimization. Classical and modern zero-sum theory owns Davenport constants, subset/weighted/universal variants, basis obstructions, and direct sums. Proof complexity and formal methods already distinguish object difficulty from lower bounds inside a restricted language.

The residual contribution is the exact production realization in quantum compilation: the same algebraic certificate is tight for R6M, loose by five-to-one for R6I, and unboundedly loose in additive budget under products. The stronger R6I transformation pinpoints the missing proof operation—whole-system Tag reconstruction—rather than merely reporting a smaller number.

8. Reproducibility

The production R6I alphabets, basis witnesses, source/generic/native agreement, and intrinsic support-one parent are commit-bound in the B1 package. The R6M all-size upper theorem and its exact support-one obstruction witness are commit-bound in the A1 parent package at `research/extensions/orion-qg/paper_a_a1_multitag_tare.py` and `research/extensions/orion-qg/PAPER_A_A1_MULTITAG_TARE_RESULTS_2026-08-24.json`; that package records `multitag_sharpness_authority: false`, so it binds the `kappa_M = 2` boundary corollary and does not carry general multitag-sharpness authority. The R2 verifier checks alphabet invariants on small nonbinary and binary groups, basis obstructions through dimension four, and the exact product formulas.

All-size authority comes from Theorems 1 and 3. Internal independent implementations remain distinct from external replication.

9. Limitations

1. The abstract exactness theorem concerns the stated zero-sum deletion language.
2. A compiler lower bound requires realization of the zero-sum-free witness.
3. No lower bound is proved for every local, syndrome-preserving, or unrestricted proof system.
4. The product amplifies one mechanism and forbids cross-component transformations by definition.
5. $\Theta(n^{(4t)})$ concerns a direct support enumerator, not arbitrary algorithms.
6. Structural support is not a physical quantum-resource advantage.
7. Submission-date novelty and independent proof review remain external.

10. Discussion and conclusion

A certificate can be sound, useful, and internally optimal while measuring the wrong layer for an intrinsic interpretation. $\mathbf{zsf}(H;A)$ identifies the exact expressive ceiling of zero-sum deletion. R6M shows that this ceiling can coincide with intrinsic support. R6I shows that a stronger global transformation can cross it dramatically.

The practical reporting rule is simple: label a number as intrinsic only after a compiler lower witness; otherwise state the normalization or proof system that owns it. This turns later improvements from apparent contradictions into interpretable increases in proof-language expressivity.

Selected references

- I. Aliev et al., *The Support of Integer Optimal Solutions*, SIAM J. Optim. 28, 2152–2157 (2018), DOI 10.1137/17M1162792.
- G. Wang, *The universal zero-sum invariant and weighted zero-sum for infinite abelian groups*, Commun. Algebra 53 (2025), DOI 10.1080/00927872.2024.2418017.
- M. Freeze and W. A. Schmid, *Remarks on a generalization of the Davenport constant*, Discrete Math. 310, 3373–3389 (2010), arXiv:0905.4248.

Publication decision record

Primary target: Quantum, provided the introduction makes the compiler/proof-language consequence immediate and the reproducibility archive is external-review ready. **Stretch:** PRX Quantum only under an independently endorsed exceptional-connection case. **R2 status:** HIGH_SELECTIVITY_SPECIALIST_CANDIDATE__EXACT_TIGHT_VS_LOOSE_CONTROLS. **External-only gates:** hostile proof audit, full primary-source overlap check, figures, exact formatting, permanent archive.